



Report 3

Build and Validate a Layer 2 Lab Network

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**NETWORK SECURITY –
BATCH 2**



Introduction:

The objective of this task was to build and validate a Layer 2 network using Cisco Packet Tracer. Three Cisco 2960 switches and six PCs were connected to create a secure switched network. Four VLANs were configured to logically separate the departments, trunk links were established between switches, ARP communication was analysed, and Sticky MAC Port Security was implemented to enhance network security.

Objectives:

- Build a Layer 2 network using three switches and six hosts.
- Create and configure four VLANs.
- Configure trunk links between switches.
- Capture and analyze ARP request and reply to packets.
- Configure Sticky MAC Port Security.
- Verify VLANs, MAC address table, and port security.

Software and Tools Used:

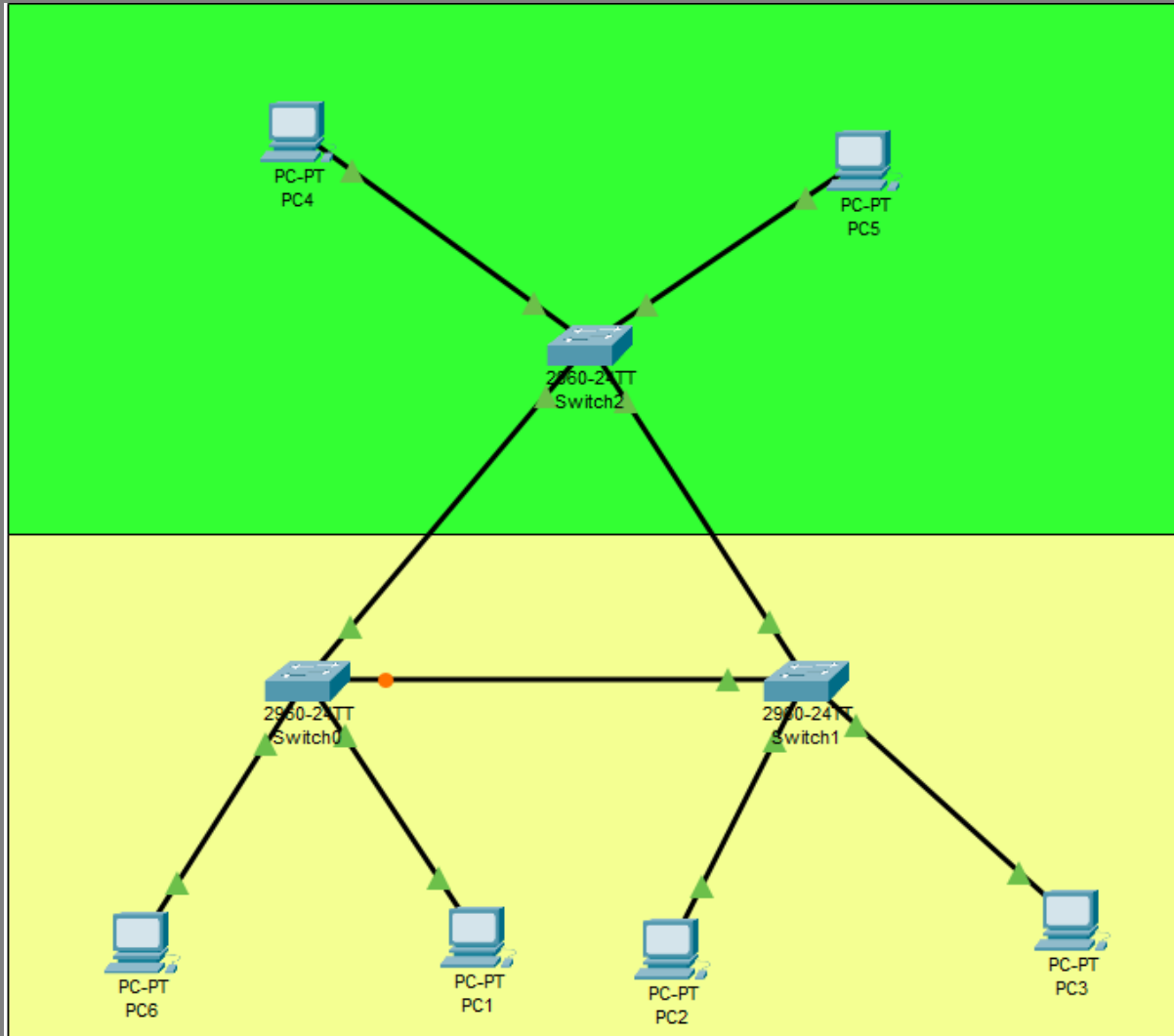
- Cisco Packet Tracer
- Cisco 2960 Switches
- Six PCs
- Simulation Mode

VLAN Configuration:

VLAN ID	VLAN Name
10	HR
20	IT
30	Admin
40	Guest

Network Topology:

A Layer 2 topology consisting of three Cisco 2960 switches and six PCs was created. Trunk links were configured between the switches to carry VLAN traffic, while each PC was connected to an access port assigned to its respective VLAN.



Access ports were assigned to their respective VLANs, and trunk ports were configured to allow VLAN traffic between switches.

VLAN Creation:

This screenshot shows the creation of VLAN 10 (HR), VLAN 20 (IT), VLAN 30 (Admin), and VLAN 40 (Guest) on the Cisco switches.

```
Switch>
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name GUEST
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#wr
Building configuration...
[OK]
Switch#
```

On Switch 2:

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name GUEST
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#wr
Building configuration...
[OK]
Switch#
```

On Switch 3:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name GUEST
Switch(config-vlan)#end
Switch#
```

Access Port Configuration:

This screenshot shows the configuration of access ports assigned to their respective VLANs. Each connected PC was placed in the correct VLAN according to the network design.

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
                                     ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access vlan 20
                                     ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 20
```

Switch 2:

```
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#int f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#int f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 40
Switch(config-if)#end
Switch#
%SYS-S-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

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Switch 3:

Physical	Config	CLI	Attributes
----------	--------	-----	------------

IOS Command Line Interface

```
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 10
Switch(config-if)#int f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
```

Trunk Port Configuration:

This screenshot shows the configuration of trunk ports between the switches using IEEE 802.1Q trunking. The trunk links allow all VLAN traffic to pass between the switches.

Switch 1:

```
Switch(config-if)#int f0/23
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

Switch(config-if)#int f0/24
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch(config-if)#en
% Ambiguous command: "en"
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

Switch 2:

```
Switch(config-if)#int f0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#int f0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

wr
Building configuration...
[OK]
Switch#
```

Switch 3:

```
Switch(config-if)#int f0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#int f0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

wr
```

VLAN Verification:

The output confirms that all VLANs were successfully created and that the switch ports were assigned to the correct VLANs.

Switch 1:

```
Switch#
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Gig0/1, Gig0/2
10 HR	active	Fa0/1
20 IT	active	Fa0/2
30 ADMIN	active	
40 GUEST	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Switch 2:

```
Switch#
Switch#
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Gig0/1, Gig0/2
10 HR	active	
20 IT	active	
30 ADMIN	active	Fa0/1
40 GUEST	active	Fa0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Switch 3:

```
Switch#
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Gig0/1, Gig0/2
10 HR	active	Fa0/1
20 IT	active	Fa0/2
30 ADMIN	active	
40 GUEST	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

MAC Address Table Verification:

The MAC address table was verified using the show mac address-table command. After communication between hosts, the switches dynamically learned the MAC addresses of connected devices.

```
Switch#show mac address-table
Mac Address Table
-----
```

Vlan	Mac Address	Type	Ports
1	0007.ecc4.0518	DYNAMIC	Fa0/24
10	0007.ecc4.0518	DYNAMIC	Fa0/24
20	0007.ecc4.0518	DYNAMIC	Fa0/24
30	0007.ecc4.0518	DYNAMIC	Fa0/24
40	0007.ecc4.0518	DYNAMIC	Fa0/24

```
Switch#
```

☐ Top

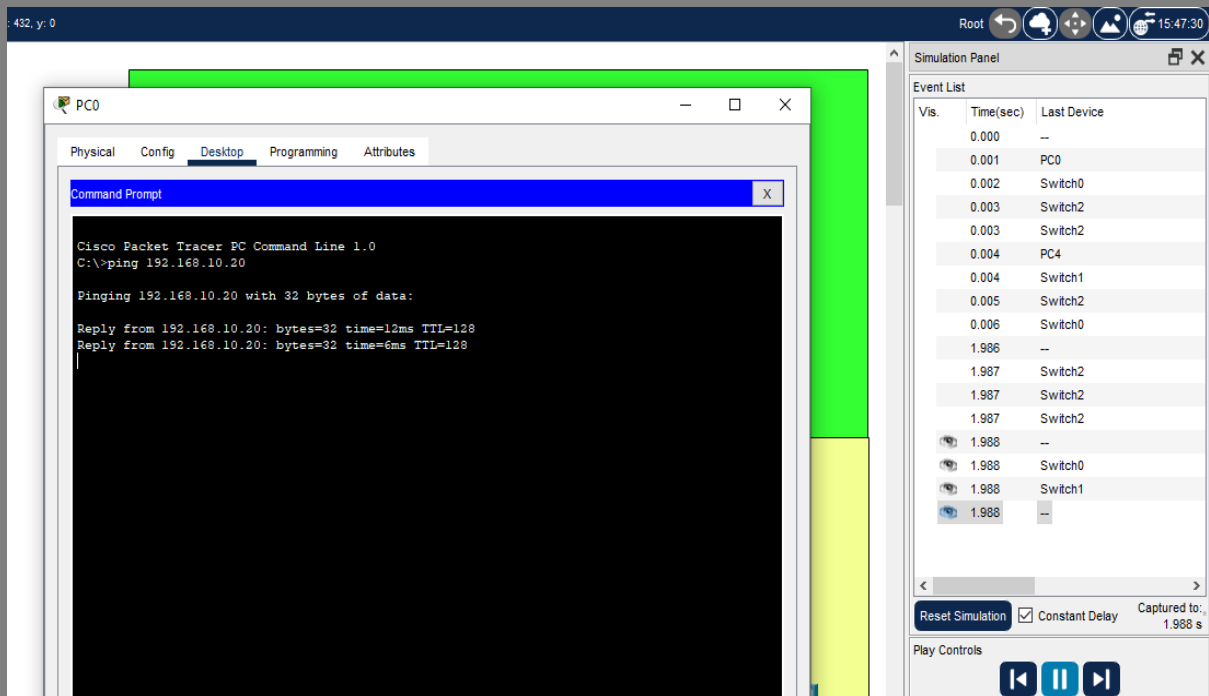
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ARP Request and Reply Analysis:

The simulation shows the ARP Request being broadcast by the source host to discover the destination MAC address. The destination host responds with an ARP Reply containing its MAC address. This confirms that ARP resolution is functioning correctly within the Layer 2 network.

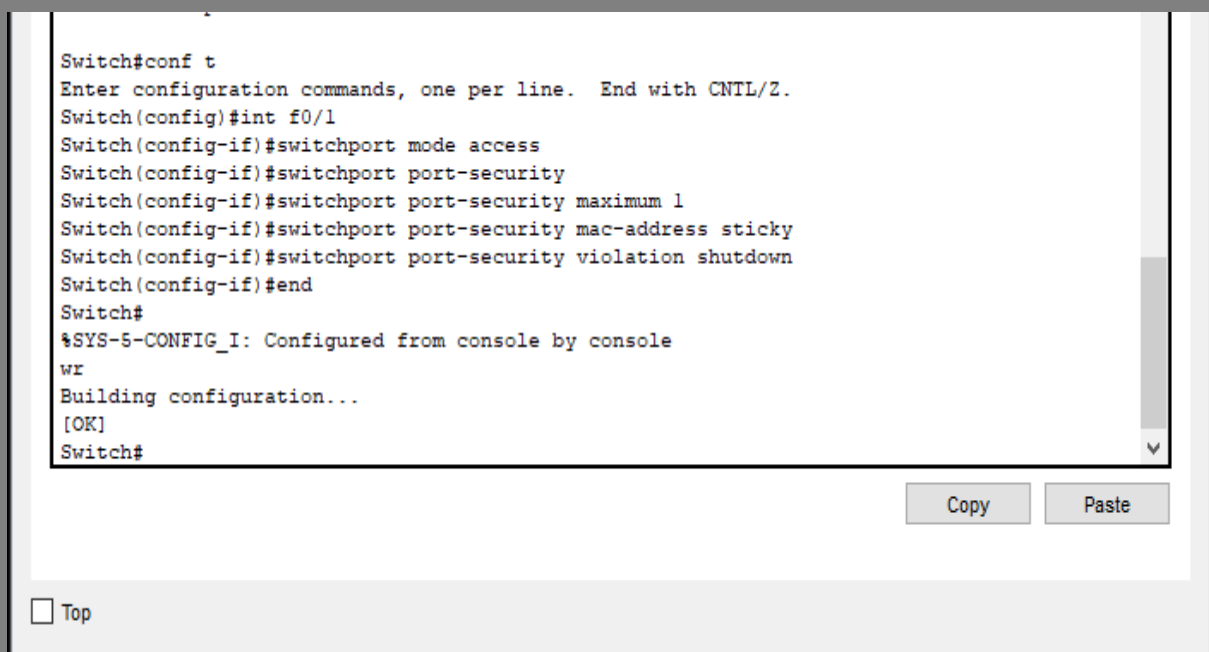
Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.002	Switch0	Switch2	ARP
	0.003	Switch2	PC4	ARP
	0.003	Switch2	Switch1	ARP
	0.004	PC4	Switch2	ARP
	0.004	Switch1	Switch0	ARP
	0.005	Switch2	Switch0	ARP
	0.006	Switch0	PC0	ARP

Cisco Packet Tracer Simulation Mode was used to observe ARP communication.



Sticky MAC Port Security:

Sticky MAC Port Security was configured on the access ports to allow only authorized devices to connect.



The screenshot shows the configuration of Sticky MAC Port Security, where the switch automatically learns and stores the MAC address of the connected device.

Port Security Verification:

The output of show port-security interface confirms that port security is enabled, the maximum allowed MAC address is configured, and Sticky MAC learning is active.

```
Switch#show port-security int f0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#
```

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The output confirms that Port Security is enabled on FastEthernet0/1 with a maximum of one secure MAC address. The switch has successfully learned one sticky MAC address, and the Security Violation count is 1, indicating that an unauthorized device attempted to connect to the secured port.

```
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
              (Count)          (Count)          (Count)
-----
Fa0/1         1             1             1             Shutdown
-----
Switch#
```

Sticky MAC Address Table:

The output of show port-security address displays the learned sticky MAC addresses stored on the switch, verifying that only authorized devices are permitted on the secured port.

```
Switch#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
-----
10      0090.2173.82D8   SecureSticky        Fa0/1    -
-----
```

Interface Status Before Port Security Violation:

This screenshot shows the interface status before testing Sticky MAC Port Security. Ports Fa0/1 and Fa0/2 are in the connected state and are assigned to VLAN 10 and VLAN 20, respectively. The trunk ports (Fa0/23 and Fa0/24) are also active, confirming that the switch is operating normally before any security violation occurs.

```
Switch#show int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	10	auto	auto	10/100BaseTX
Fa0/2		connected	20	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		connected	trunk	auto	auto	10/100BaseTX
Fa0/24		connected	trunk	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100BaseTX
Gig0/2		notconnect	1	auto	auto	10/100BaseTX

Interface Status After Connecting an Unauthorized Device:

This screenshot shows the interface status after connecting a different (unauthorized) device to the secured port. Due to the Sticky MAC Port Security configuration, Fa0/1 entered the err-disabled state. This indicates that a port security violation occurred because the new device's MAC address did not match the learned sticky MAC address.

```
Switch#show int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		err-disabled	10	auto	auto	10/100BaseTX
Fa0/2		connected	20	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX

--More--

Spanning Tree Protocol (STP) Verification:

The output of the show spanning-tree command confirms that STP is enabled and operating correctly. It displays the root bridge information, the bridge priority, VLAN details, and the role of each switch port (Root Port, Designated Port, or Alternate/Blocking Port). This verifies that the network has a loop-free topology and that redundant links are being managed correctly by STP.

```
VLAN0010
Spanning tree enabled protocol ieee
Root ID    Priority    32778
           Address    0000.0C01.EAEB
           Cost       19
           Port       24(FastEthernet0/24)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
           Address    00D0.BAE6.6C60
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/22	Altn	BLK	19	128.22	P2p
Fa0/24	Root	FWD	19	128.24	P2p

```
VLAN0020
Spanning tree enabled protocol ieee
Root ID    Priority    32788
           Address    0000.0C01.EAEB
           Cost       19
           Port       24(FastEthernet0/24)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
           Address    00D0.BAE6.6C60
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/22	Altn	BLK	19	128.22	P2p
Fa0/24	Root	FWD	19	128.24	P2p

```
VLAN0030
Spanning tree enabled protocol ieee
Root ID    Priority    32798
           Address    0000.0C01.EAEB
           Cost       19
           Port       24(FastEthernet0/24)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32798 (priority 32768 sys-id-ext 30)
           Address    00D0.BAE6.6C60
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/22	Altn	BLK	19	128.22	P2p
Fa0/24	Root	FWD	19	128.24	P2p

Vlan 40:

```
VLAN0040
Spanning tree enabled protocol ieee
Root ID    Priority    32808
           Address    0000.0C01.EAEB
           Cost       19
           Port       24(FastEthernet0/24)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32808  (priority 32768 sys-id-ext 40)
           Address    00D0.BAE6.6C60
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/22         Altn BLK 19      128.22  P2p
Fa0/24         Root FWD 19      128.24  P2p
```

Business Justification:

Sticky MAC Port Security helps reduce business risks by allowing only authorized devices to connect to the network. It prevents unauthorized access, protects sensitive organizational data, and minimizes the risk of MAC spoofing and insider attacks. This improves the overall security and reliability of the enterprise network.

Learning Outcomes:

Through this task, I gained practical experience in creating VLANs, configuring trunk links, verifying MAC address learning, analysing ARP communication, and implementing Sticky MAC Port Security. I also learned how Layer 2 security mechanisms help protect enterprise networks from unauthorized access.

Conclusion:

The Layer 2 network was successfully built and validated using Cisco Packet Tracer. VLANs, trunk links, ARP communication, MAC address learning, and Sticky MAC Port Security were successfully configured and verified. The task enhanced my understanding of Layer 2 switching technologies and network security best practices.